Year 11 Foundation

REVISION PACK

Progress Check 3

Thursday 12 November 2026

Name: _____

Class: _____

Units in this pack

- Number 9
- Algebra 8

Number 9 · Percentages of amounts (Lesson 6.1)

Worked example: a percentage of an amount

Work out 35% of 240 KD.

Step 1	Find 10%	$240 \div 10 = 24$
Step 2	Find 30%	$3 \times 24 = 72$
Step 3	Find 5%	$24 \div 2 = 12$
Step 4	Answer: 30% + 5%	$72 + 12 = 84\ \text{KD}$

Practice: a percentage of an amount

1.1 Work out 15% of 80 KD.

1.2 Work out 65% of 360 KD.

1.3 Work out 12% of 650 KD.

1.4 Work out 25% of 180 KD.

1.5 Work out 5% of 340 KD.

1.6 Work out 75% of 96 KD.

Number 9 · Percentages of amounts (Lesson 6.1)

Worked example: writing one number as a percentage of another

In a test, Ali scores 42 marks out of 60. Write 42 out of 60 as a percentage.

Step 1	Make a fraction	$\frac{42}{60}$
Step 2	Multiply by 100	$\frac{42}{60} \times 100 = 70$
Step 3	Answer	70%

Practice: writing one number as a percentage of another

1.7 In a test, Reem scores 18 marks out of 25. Write 18 out of 25 as a percentage.

1.8 In a test, Omar scores 39 marks out of 60. Write 39 out of 60 as a percentage.

1.9 In a test, Noor scores 63 marks out of 84. Write 63 out of 84 as a percentage.

1.10 In a test, Bader scores 27 marks out of 30. Write 27 out of 30 as a percentage.

1.11 In a test, Maha scores 21 marks out of 35. Write 21 out of 35 as a percentage.

1.12 In a test, Fahad scores 56 marks out of 80. Write 56 out of 80 as a percentage.

Number 9 · Percentages of amounts (Lesson 6.1)

Worked example: a percentage of the rest

Ali earns 1000 KD a month. Ali saves 10% of it. Ali spends 20% of the rest on food. How much does Ali spend on food?

Step 1	Saves 10%	10% of $1000 = 100$
Step 2	The rest	$1000 - 100 = 900$
Step 3	20% of the rest	$0.2 \times 900 = 180$
Step 4	Answer	180 KD

Practice: a percentage of the rest

1.13 Mona earns 1000 KD a month. Mona saves 20% of it. Mona spends 30% of the rest on rent. How much does Mona spend on rent?

1.14 Omar earns 1500 KD a month. Omar saves 10% of it. Omar spends 40% of the rest on rent. How much does Omar spend on rent?

1.15 Dana earns 800 KD a month. Dana saves 25% of it. Dana spends 50% of the rest on food. How much does Dana spend on food?

1.16 Yousef earns 2000 KD a month. Yousef saves 20% of it. Yousef spends 25% of the rest on food. How much does Yousef spend on food?

1.17 Reem earns 1200 KD a month. Reem saves 25% of it. Reem spends 20% of the rest on rent. How much does Reem spend on rent?

1.18 Khalid earns 900 KD a month. Khalid saves 10% of it. Khalid spends 50% of the rest on food. How much does Khalid spend on food?

Number 9 · Percentage increase, decrease and profit (Lesson 6.2)

Worked example: increasing an amount

Increase 60 KD by 15%.

Step 1	Find 15%	$0.15 \times 60 = 9$
Step 2	Add	$60 + 9 = 69$
Step 3	Answer	$69\ \text{KD}$

Practice: increasing an amount

2.1 Increase 40 KD by 10%.

2.2 Increase 75 KD by 14%.

2.3 Increase 250 KD by 8%.

2.4 Increase 120 KD by 5%.

2.5 Increase 64 KD by 25%.

2.6 Increase 90 KD by 30%.

Number 9 · Percentage increase, decrease and profit (Lesson 6.2)

Worked example: decreasing an amount

A car is worth 5000 KD. Its value falls by 20%. Work out the new value of the car.

Step 1	Find 20%	$0.2 \times 5000 = 1000$
Step 2	Subtract	$5000 - 1000 = 4000$
Step 3	Answer	$4000\ \text{KD}$

Practice: decreasing an amount

2.7 A car is worth 3000 KD. Its value falls by 10%. Work out the new value of the car.

2.8 A car is worth 7200 KD. Its value falls by 15%. Work out the new value of the car.

2.9 A sofa costs 480 KD. In a sale the price is reduced by 25%. Work out the sale price.

2.10 A phone costs 350 KD. In a sale the price is reduced by 20%. Work out the sale price.

2.11 A car is worth 9000 KD. Its value falls by 5%. Work out the new value of the car.

2.12 A TV costs 640 KD. In a sale the price is reduced by 15%. Work out the sale price.

Number 9 · Percentage increase, decrease and profit (Lesson 6.2)

Worked example: percentage profit

A shop buys a lamp for 20 KD. It sells the lamp for 25 KD. Work out the percentage profit.

Step 1	Profit	$25 - 20 = 5$
Step 2	Divide by the cost	$5 \div 20 = 0.25$
Step 3	Multiply by 100	$0.25 \times 100 = 25$
Step 4	Answer	25%

Practice: percentage profit

2.13 A shop buys a hat for 50 KD. It sells it for 60 KD. Work out the percentage profit.

2.14 A shop buys a toy for 40 KD. It sells it for 46 KD. Work out the percentage profit.

2.15 A shop buys a chair for 80 KD. It sells it for 100 KD. Work out the percentage profit.

2.16 A shop buys a kettle for 60 KD. It sells it for 78 KD. Work out the percentage profit.

2.17 A shop buys a bike for 200 KD. It sells it for 220 KD. Work out the percentage profit.

2.18 A shop buys a book for 16 KD. It sells it for 20 KD. Work out the percentage profit.

Number 9 · Reverse percentages (Lesson 6.4)

Worked example: price before an increase

After a 10% increase, the price of a game is 66 KD. Work out the price before the increase.

Step 1	New price is 110%	$110\% = 66$
Step 2	Find 1%	$66 \div 110 = 0.6$
Step 3	Find 100%	$0.6 \times 100 = 60$
Step 4	Answer	$60\ \text{KD}$

Practice: price before an increase

3.1 After a 20% increase, the price of a clock is 72 KD. Work out the price of the clock before the increase.

3.2 After a 10% increase, the price of a game is 55 KD. Work out the price before the increase.

3.3 After a 30% increase, the price of a bike is 130 KD. Work out the price before the increase.

3.4 After a 5% increase, the price of a fan is 84 KD. Work out the price before the increase.

3.5 After a 25% increase, the price of a heater is 150 KD. Work out the price before the increase.

3.6 After a 15% increase, the price of a lamp is 46 KD. Work out the price before the increase.

Number 9 · Reverse percentages (Lesson 6.4)

Worked example: price before a reduction

In a sale, all prices are reduced by 20%. The sale price of a coat is 48 KD. Work out the price before the sale.

Step 1	Sale price is 80%	$80\% = 48$
Step 2	Find 1%	$48 \div 80 = 0.6$
Step 3	Find 100%	$0.6 \times 100 = 60$
Step 4	Answer	$60\ \text{KD}$

Practice: price before a reduction

3.7 In a sale, all prices are reduced by 25%. The sale price of a skirt is 36 KD. Work out the price of the skirt before the sale.

3.8 In a sale, all prices are reduced by 35%. The sale price of a jacket is 52 KD. Work out the price before the sale.

3.9 In a sale, all prices are reduced by 10%. The sale price of a lamp is 63 KD. Work out the price before the sale.

3.10 In a sale, all prices are reduced by 10%. The sale price of a shirt is 45 KD. Work out the price before the sale.

3.11 In a sale, all prices are reduced by 20%. The sale price of a bag is 56 KD. Work out the price before the sale.

3.12 In a sale, all prices are reduced by 40%. The sale price of a table is 120 KD. Work out the price before the sale.

Algebra 8 · Simultaneous equations (Lesson 12.1)

Worked example: multiplying both equations

Solve the simultaneous equations $3x + 2y = 16$ and $2x + 3y = 14$

Step 1	Make the y the same	$9x + 6y = 48, \quad 4x + 6y = 28$
Step 2	Subtract	$5x = 20$
Step 3	Find x	$x = 4$
Step 4	Put x in	$3(4) + 2y = 16$
Step 5	Solve	$y = 2$
Step 6	Answer	$x = 4,\ y = 2$

Practice: multiplying both equations

4.1 Solve the simultaneous equations

$3x + 2y = 14$

$2x + 3y = 11$

4.2 Solve the simultaneous equations

$5x + 2y = 9$

$3x + 4y = 11$

4.3 Solve the simultaneous equations

$2x + 5y = 11$

$3x + 2y = 11$

4.4 Solve the simultaneous equations

$4x + 3y = 18$

$3x + 2y = 13$

4.5 Solve the simultaneous equations

$2x + 3y = 14$

$5x + 4y = 21$

4.6 Solve the simultaneous equations

$3x + 4y = 10$

$2x + 5y = 9$

Algebra 8 · Simultaneous equations (Lesson 12.1)

Worked example: different signs

Solve the simultaneous equations $4x + 3y = 10$ and $2x - y = 0$

Step 1	Make the y the same	$2x - y = 0 \xrightarrow{\times 3} 6x - 3y = 0$
Step 2	Different signs: add	$10x = 10$
Step 3	Find x	$x = 1$
Step 4	Put x in	$2(1) - y = 0$
Step 5	Solve	$y = 2$
Step 6	Answer	$x = 1,\ y = 2$

Practice: different signs

4.7 Solve the simultaneous equations

$3x + 2y = 9$

$4x - y = 1$

4.8 Solve the simultaneous equations

$5x + 3y = 7$

$2x - y = 5$

4.9 Solve the simultaneous equations

$2x + 3y = 9$

$5x - 2y = 13$

4.10 Solve the simultaneous equations

$3x + 2y = 8$

$5x - 4y = 6$

4.11 Solve the simultaneous equations

$4x + y = 6$

$3x - 2y = -1$

4.12 Solve the simultaneous equations

$2x + 5y = 3$

$3x - y = 13$

Algebra 8 · Simultaneous equations (Lesson 12.1)

Worked example: word problems

2 pens and 3 rulers cost 1.300 KD. 3 pens and 1 ruler cost 1.250 KD. Work out the cost of one pen and the cost of one ruler.

Step 1	Use letters	$2p + 3r = 1.3, \quad 3p + r = 1.25$
Step 2	Multiply by 3	$9p + 3r = 3.75$
Step 3	Subtract	$7p = 2.45$
Step 4	Divide by 7	$p = 0.35$
Step 5	Find r	$r = 1.25 - 3(0.35) = 0.2$
Step 6	Answer	$\text{pen } 0.350 \text{ KD}, \ \text{ruler } 0.200 \text{ KD}$

Practice: word problems

4.13 2 wraps and 3 teas cost 3.000 KD. 3 wraps and 1 tea cost 2.750 KD. Work out the cost of one wrap and the cost of one tea.

4.14 4 pens and 1 notebook cost 2.250 KD. 2 pens and 3 notebooks cost 1.750 KD. Work out the cost of one pen and the cost of one notebook.

4.15 3 sandwiches and 2 coffees cost 6.000 KD. 2 sandwiches and 5 coffees cost 6.750 KD. Work out the cost of one sandwich and the cost of one coffee.

4.16 2 cakes and 1 water cost 3.000 KD. 1 cake and 3 waters cost 2.750 KD. Work out the cost of one cake and the cost of one water.

4.17 3 apples and 4 oranges cost 3.000 KD. 2 apples and 1 orange cost 1.500 KD. Work out the cost of one apple and the cost of one orange.

4.18 5 pencils and 2 erasers cost 3.400 KD. 3 pencils and 4 erasers cost 4.000 KD. Work out the cost of one pencil and the cost of one eraser.